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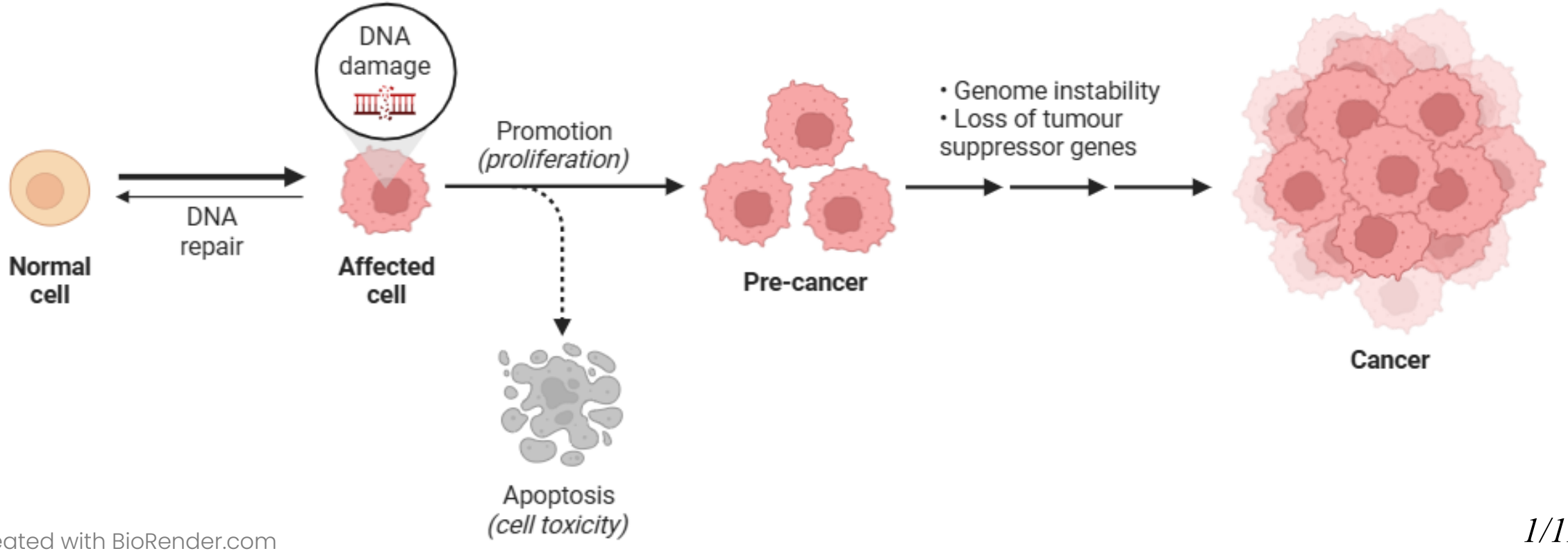
BREAST TUMOUR CLASSIFICATION FROM MORPHOLOGICAL FEATURES

Machine learning pipeline using the **Breast Cancer Wisconsin** dataset.

BREAST CANCER: BACKGROUND

- most commonly diagnosed cancer in women worldwide
- about 2.3 million new cases and 670,000 deaths annually (Freihat et al., 2025)
- diagnosis often relies on imaging and morphological features → research question



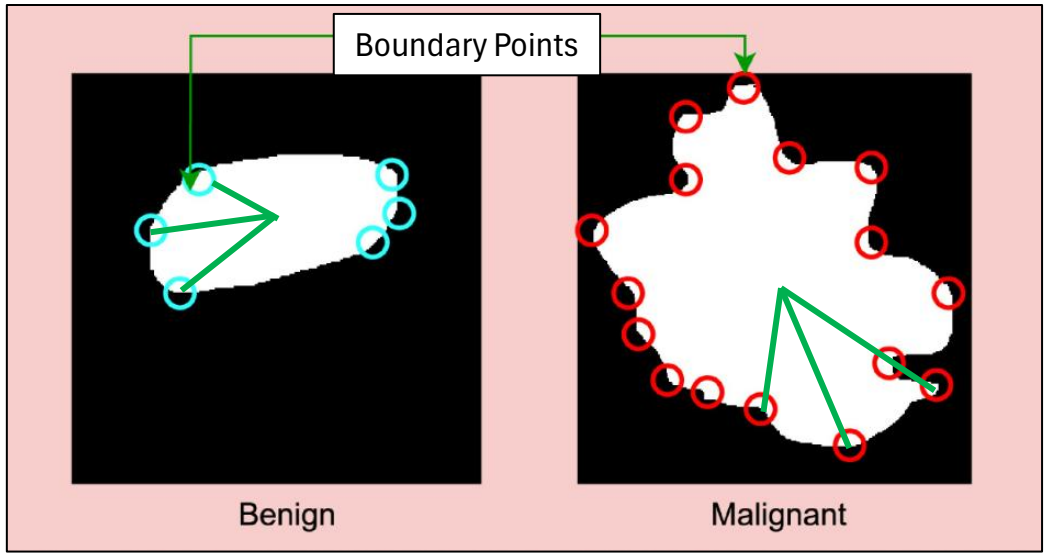
Created with BioRender.com

Research Question: Can tumour morphology features predict breast tumour malignancy?

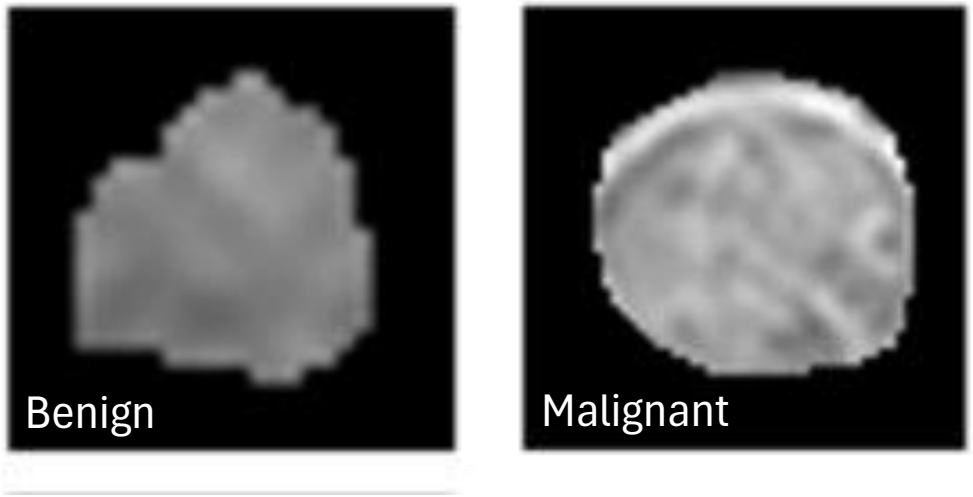
BIOLOGY → DATA → MODELS → INSIGHT

The **Breast Cancer Wisconsin (Diagnostic)** Dataset (Wolberg et al., 1995) looks at various morphological features of benign and malignant tumours.

We selected a subset of interpretable morphological features for rapid prototyping:



average radius



average texture

Montaha et al., 2024 (left); Alksas et al., 2021 (right)

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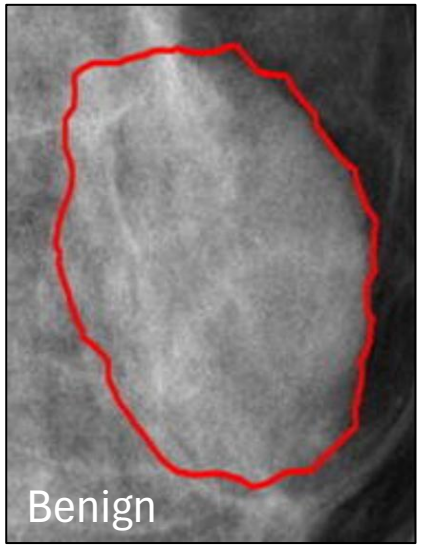
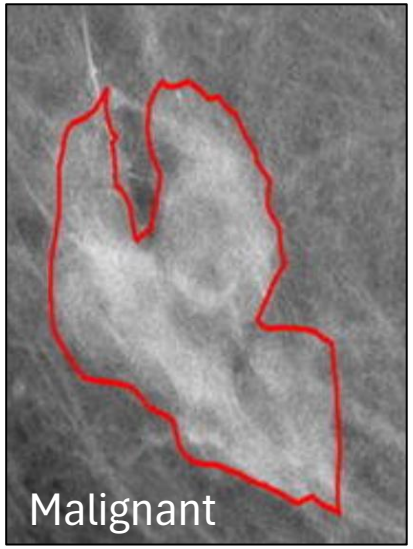
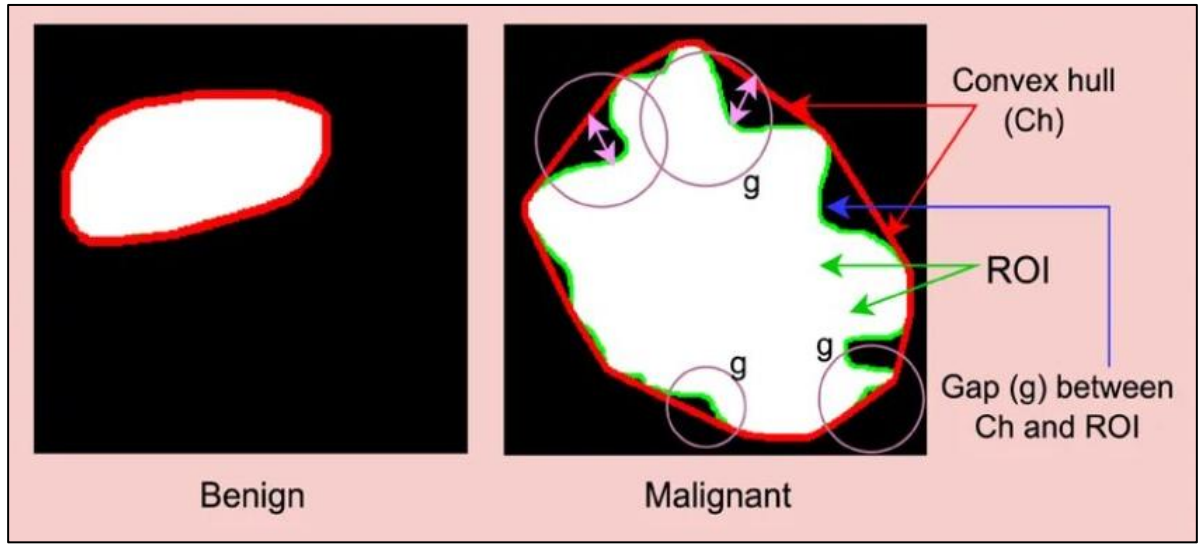




average smoothness



average concavity

Li et al., 2017 (left); Montaha et al., 2024 (right)

The **Breast Cancer Wisconsin (Diagnostic)** Dataset (Wolberg et al., 1995) looks at various morphological features of benign and malignant tumours.

We selected a subset of interpretable morphological features for rapid prototyping:

Feature	Unit	Low value → High value
average radius	unitless	small → large
average texture	unitless	uniform → heterogenous
average smoothness	unitless	smooth → irregular
average concavity	unitless	convex → concave

METHODS

① ***Exploratory Data Analysis (R)*** **:** *Distribution analysis, Class separability*

Are the benign and malignant classes **separable**?

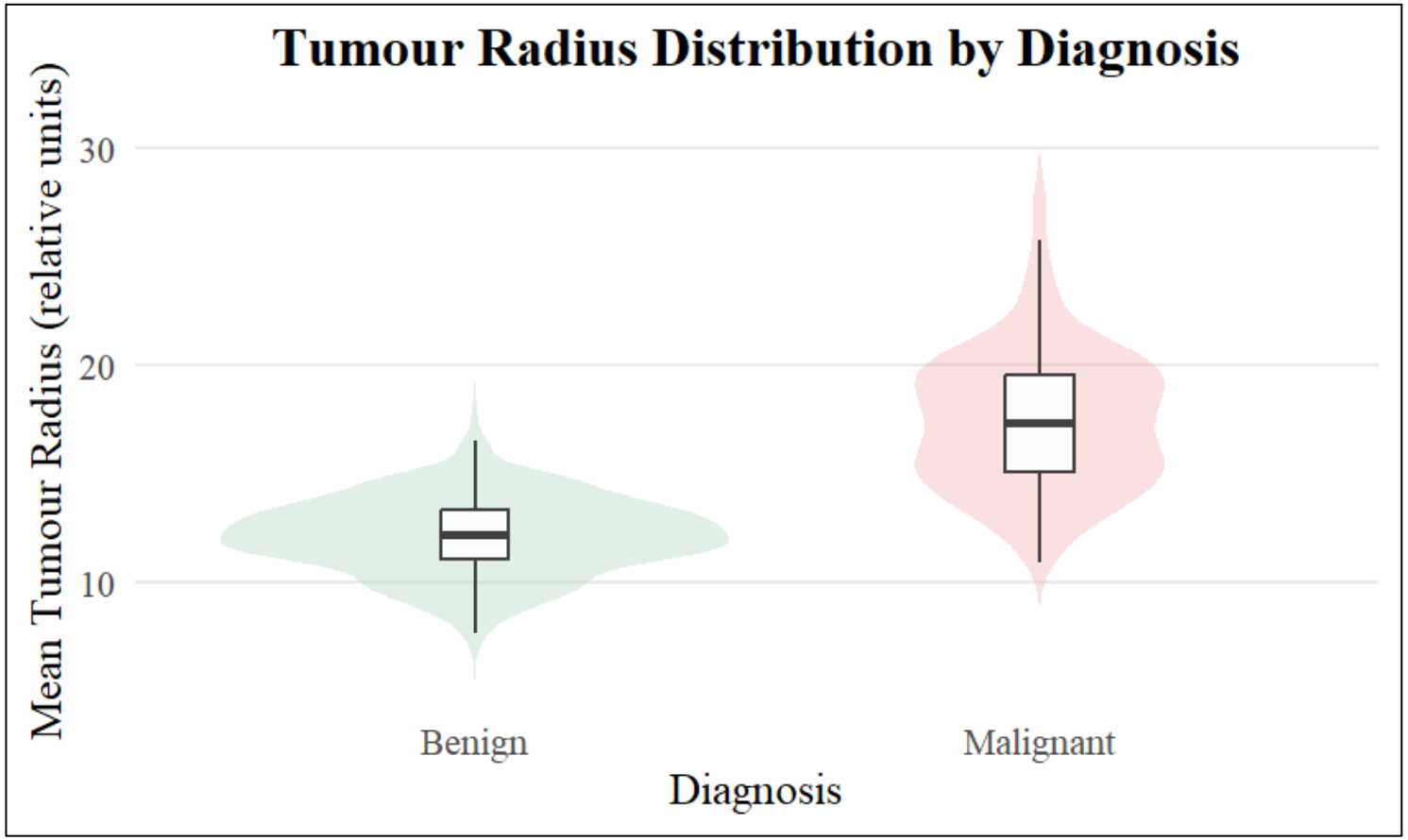
Can we train a model to **predict** a benign/malignant tumour?

② ***ML Model Development (Python)*** **:** *Logistic regression, Train/test split, Feature scaling*

③ ***Model Validation (Python)*** **:** *ROC -AUC , Classification performance*

How **accurate** is the model?

1. EXPLORATORY DATA ANALYSIS



Violin plot in R shows that benign and malignant classes are separable.

2. MODEL DEVELOPMENT

```python
# 7. Morphology Simulator

# --- input values ---

example_tumours = pd.DataFrame({
    "Description": ["Small, smooth tumour", "Large, concave tumour"],
    "radius_mean": [8, 18],
    "texture_mean": [12, 20],
    "smoothness_mean": [0.12, 0.08],
    "concavity_mean": [0.02, 0.18]
})

example_tumours["Malignancy Probability"] = example_tumours.apply(
    lambda row: simulate_prediction(
        row["radius_mean"], row["texture_mean"],
        row["smoothness_mean"], row["concavity_mean"]
    ),
    axis=1
)

print("\nMorphology Simulation Results:")
print(example_tumours)
```

```
Morphology Simulation Results:
             Description  radius_mean  texture_mean  smoothness_mean  concavity_mean  Malignancy Probability
0   Small, smooth tumour            8            10             0.11            0.03                0.000128
1  Large, concave tumour           19            18             0.06            0.21                0.903820
```

Video demonstration: simulating tumour morphology inputs to estimate malignancy probability.

Inputs/Outputs:

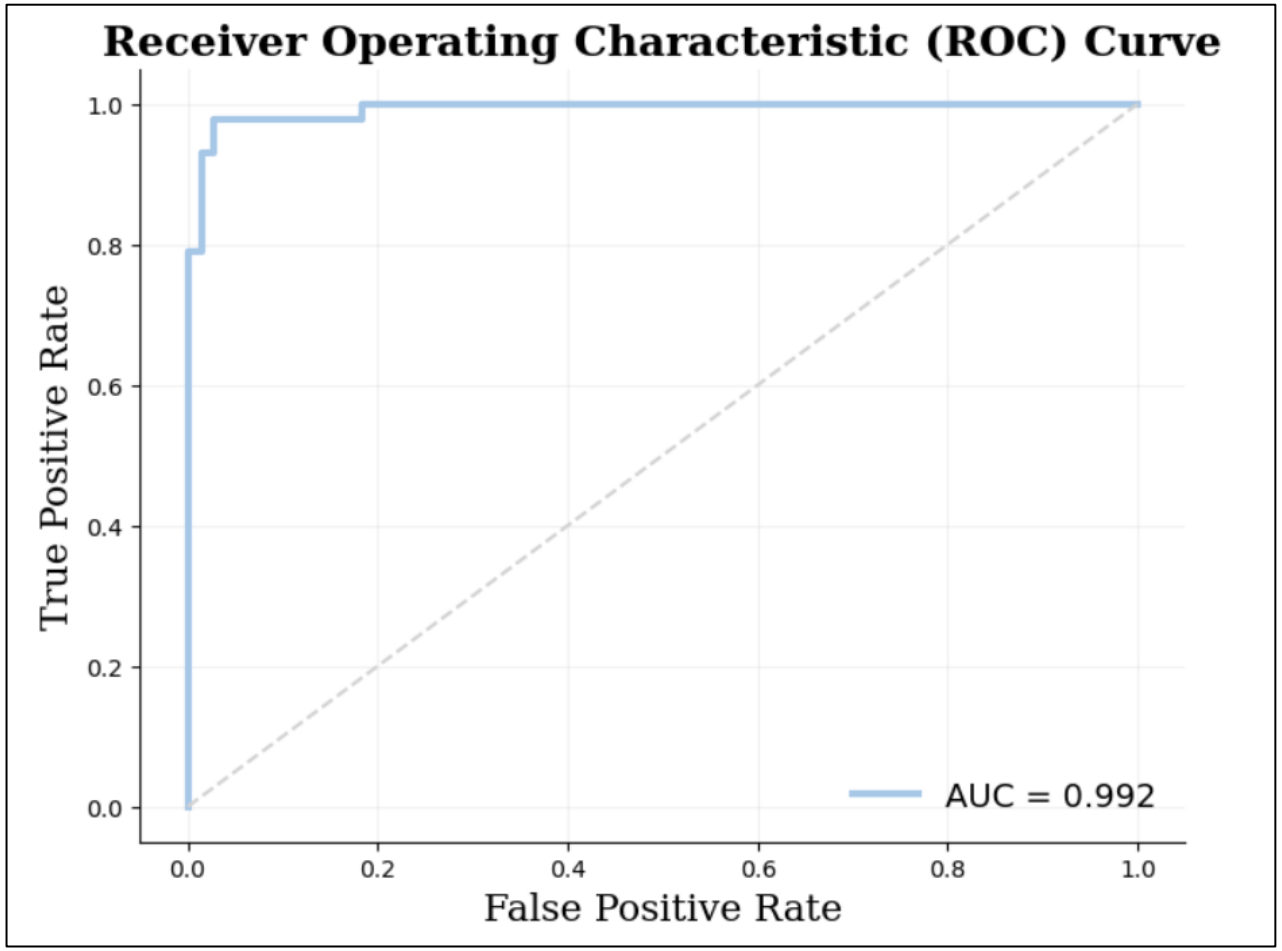
```
"radius_mean": [8, 8],
"texture_mean": [11, 11],
"smoothness_mean": [0.12, 0.05],
"concavity_mean": [0.04, 0.04]
```

```
Malignancy Probability
                0.000480
                0.000001
```

```
"radius_mean": [8, 16],
"texture_mean": [11, 19],
"smoothness_mean": [0.20, 0.04],
"concavity_mean": [0.04, 0.13]
```

```
Malignancy Probability
                0.311207
                0.037473
```

3. ML MODEL VALIDATION



Outputs:

```
Model Accuracy on Test Set: 0.9649122807017544

Feature Coefficients (importance):
radius_mean: 3.2381
texture_mean: 1.2857
smoothness_mean: 1.1901
concavity_mean: 1.3159


AUC: 0.9924664264657714
```

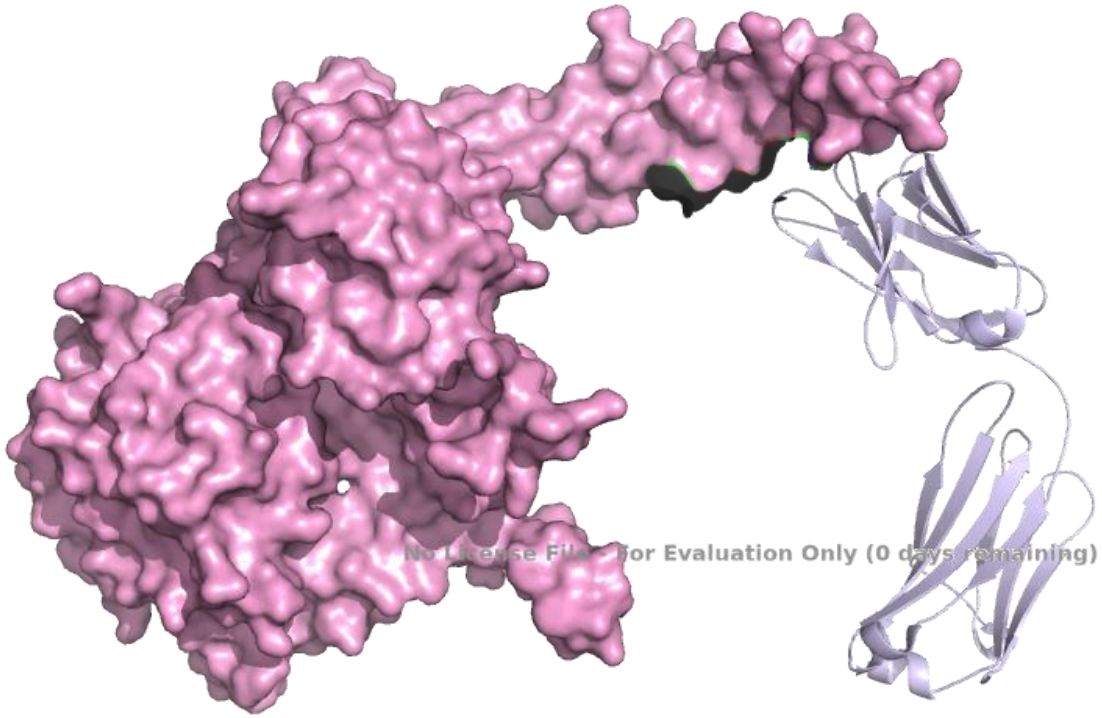
High AUC (0.99)
indicates excellent discrimination between benign and malignant tumours.

KEY LIMITATIONS

- Small, curated dataset (limited generalizability)
- Limited feature set (4 selected variables) due to time constraints
- No external validation (model solely tested on the same dataset from which it originated)
- Not representative of real clinical deployment
- Does not replace clinical judgment

FUTURE INSIGHTS & RESEARCH

- Model expansion with additional features
- Educational tool for students
- Clinical dashboards for interdisciplinary teams
- Integration of molecular data
- Application to new datasets
- **Long-term goal: build a validated diagnostic decision-support tool**



HER2 receptor (purple), a well-known clinical target in breast cancer, bound to a therapeutic antibody fragment (pink). Retrieved from Protein Data Bank (PDB 1D: 1N8Z) and generated in PYMOL.

References

Alksas, A., Shehata, M., & Saleh, G. (2021). A novel computer-aided diagnostic system for accurate detection and grading of liver tumors.Scientific Reports. https://doi.org/10.1038/s41598-021-91634-0

Freihat, O., Sipos, D., & Kovacs, A. (2025). Global burden and projections of breast cancer incidence and mortality to 2050: a comprehensive analysis of GLOBOCAN data. Frontiers in Public Health, 13, 1622954. https://doi.org/10.3389/fpubh.2025.1622954

Li, H., Meng, X., Wang, T., Tang, Y., & Yin, Y. (2017). Breast masses in mammography classification with local contour features. BioMedicalEngineering OnLine, 16(1), 44. https://doi.org/10.1186/s12938-017-0332-0

Montaha, S., Azam, S., & Bhuiyan, R. (2024). Malignancy pattern analysis of breast ultrasound images using clinical features and a graphconvolutional network. Digital Health, 10(23). https://doi.org/10.1177/20552076241251660

RCSB Protein Data Bank. (2003). *Crystal structure of extracellular domain of human HER2 complexed with Herceptin Fab* (PDB ID:1N8Z). https://doi.org/10.2210/pdb1N8Z/pdb

Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine LearningRepository. https://doi.org/10.24432/C5DW2B.